Lecture 2

Lagrange, Galois, and reduced surds

Which continued fractions repeat?

$$x > 1, \qquad -1 < x' < 0$$

Recall: the complete quotient

$$x_0 = x, \qquad a_n = \lfloor x_n \rfloor, \qquad x_{n+1} = \frac{1}{x_n - a_n}$$

The number $x_n = [a_n; a_{n+1}, \ldots]$ is the remaining tail.

$$x = [a_0; a_1, \ldots, a_{n-1}, x_n].$$

For $n \geq 1$,

$$x_n > 1, \qquad a_n \geq 1.$$

The target theorem

Lagrange: the direction still to prove

$$\text{quadratic irrational} \implies \text{eventually periodic}$$

Galois

$$\text{purely periodic} \iff x > 1,\ -1 < x' < 0$$

The second statement concerns quadratic irrationals.

We will use integer states and a reversible step.

Conjugates follow the same algebra

$$x_{n+1} = \frac{1}{x_n - a_n} \qquad \Longrightarrow \qquad x'_{n+1} = \frac{1}{x'_n - a_n}$$

The digit is still $a_n = \lfloor x_n \rfloor$.

We do not take the integer part of x'_n.

If $x'_n < 0$ and $a_n \geq 1$, then

$$x'_n - a_n < -1, \qquad -1 < x'_{n+1} < 0.$$

Reduced tails stay reduced

Suppose

$$x_n > 1, \qquad -1 < x_n' < 0.$$

Then

$$x_{n+1} > 1, \qquad -1 < x_{n+1}' < 0.$$

Once reduced, every later complete quotient is reduced.

We still have to show that an arbitrary quadratic irrational eventually reaches this region.

Every quadratic irrational reaches a reduced tail

The inverse prefix matrix gives, for $n \geq 2$,

$$x'_n = -\frac{q_{n-2}x' - p_{n-2}}{q_{n-1}x' - p_{n-1}}.$$

The convergents tend to x, and $x' \neq x$.

For large n, numerator and denominator have the same sign.

$$x'_n < 0 \quad \text{eventually.}$$

One more step puts the conjugate in $(-1, 0)$.

An integer description of a surd

Choose a fixed positive nonsquare integer D.

$$x_n = \frac{P_n + \sqrt{D}}{Q_n}, \qquad P_n, Q_n \in \mathbb{Z}, \quad Q_n \neq 0.$$

We can choose the initial representation so that

$$Q_n \mid D - P_n^2$$

holds at every step.

D need not be squarefree. Initially, Q_n may be negative.

The exact recurrence

Set $a_n = \lfloor x_n \rfloor$ and

$$P_{n+1} = a_n Q_n - P_n.$$

Then

$$\begin{aligned} \frac{1}{x_n - a_n} &= \frac{Q_n}{\sqrt{D} - P_{n+1}} \\ &= \frac{P_{n+1} + \sqrt{D}}{(D - P_{n+1}^2)/Q_n}. \end{aligned}$$

Thus

$$Q_{n+1} = \frac{D - P_{n+1}^2}{Q_n}.$$

Why the recurrence stays integral

$$P_{n+1}^2 - P_n^2 = a_n Q_n (a_n Q_n - 2P_n).$$

So Q_n divides both $D - P_n^2$ and $D - P_{n+1}^2$.

$$Q_{n+1} = (D - P_{n+1}^2)/Q_n \in \mathbb{Z}.$$

Also,

$$D - P_{n+1}^2 = Q_n Q_{n+1}.$$

This supplies the divisibility condition for the next step.

The bounds from reducedness

For a reduced state,

$$x = \frac{P + \sqrt{D}}{Q} > 1, \qquad -1 < x' = \frac{P - \sqrt{D}}{Q} < 0.$$

Subtract:

$$x - x' = \frac{2\sqrt{D}}{Q} > 1 \quad \Longrightarrow \quad 0 < Q < 2\sqrt{D}.$$

The conjugate inequalities give

$$\sqrt{D} - Q < P < \sqrt{D} \quad \Longrightarrow \quad |P| < \sqrt{D}.$$

Finitely many states

For fixed D, only finitely many integer pairs satisfy

$$|P| < \sqrt{D}, \qquad 0 < Q < 2\sqrt{D}.$$

Every sufficiently late state lies in this finite set.

Each state determines its digit and its next state.

$$(P_i, Q_i) = (P_j, Q_j) \quad \Longrightarrow \quad a_{i+k} = a_{j+k} \quad (k \geq 0).$$

This proves the converse direction of Lagrange's theorem.

The cycle for $\sqrt{3}$

n	P_n	Q_n	x_n	a_n
0	0	1	$\sqrt{3}$	1
1	1	2	$(1+\sqrt{3})/2$	1
2	1	1	$1+\sqrt{3}$	2
3	1	2	$(1+\sqrt{3})/2$	1

$$\sqrt{3} = [1; \overline{1, 2}]$$

The initial state lies outside the reduced cycle.

Your turn: $\sqrt{6}$

Start with $D = 6$ and $(P_0, Q_0) = (0, 1)$.

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = \frac{6 - P_{n+1}^2}{Q_n}.$$

Find the first repeated state.

Read off the continued fraction of $\sqrt{6}$.

Record the triples (P_n, Q_n, a_n).

The period of $\sqrt{6}$

n	P_n	Q_n	a_n
0	0	1	2
1	2	2	2
2	2	1	4
3	2	2	2

$$\sqrt{6} = [2; \overline{2, 4}]$$

Finiteness allows an initial segment

A function on the finite set $\{0, 1\}$ can satisfy

$$F(0) = 1, \qquad F(1) = 1.$$

Starting at 0 gives

$$0,\ 1,\ 1,\ 1, \ldots$$

Finiteness guarantees eventual repetition.

To prove pure periodicity, we need to recover predecessors.

Galois's criterion

For a quadratic irrational x,

$$x \text{ is purely periodic}$$

$$\Longleftrightarrow$$

$$x > 1 \qquad \text{and} \qquad -1 < x' < 0.$$

We first prove that a reduced starting point has no preperiod.

A successor remembers its reduced predecessor

Suppose u is reduced, $a = \lfloor u \rfloor$, and

$$z = \frac{1}{u - a}.$$

Conjugating gives

$$-\frac{1}{z'} = a - u', \qquad a < -\frac{1}{z'} < a + 1.$$

Therefore

$$a = \lfloor -1/z' \rfloor, \qquad u = a + \frac{1}{z}.$$

Both the previous digit and the reduced predecessor are unique.

Rewinding a repeated state

Suppose all x_n are reduced and $x_i = x_j$ with $i < j$.

$$\begin{aligned} x_i &= x_j \\ x_{i-1} &= x_{j-1} \\ &\vdots \\ x_0 &= x_{j-i}. \end{aligned}$$

Every backward step uses the unique reduced predecessor.

The complete-quotient sequence repeats from its start.

Why a pure period must be reduced

Suppose the pure period has length ℓ.

Lecture 1 shows that x is quadratic.

All sufficiently late complete quotients are reduced.

$$x_{k\ell} = x_0 \qquad (k = 1, 2, \ldots)$$

Choose k large enough.

The starting value is itself one of those reduced tails.

A reducedness test

Which of these have purely periodic continued fractions?

$$\sqrt{2}, \qquad 1+\sqrt{3}, \qquad \frac{1+\sqrt{5}}{2}, \qquad \frac{\sqrt{2}}{2}$$

Check both inequalities:

$$x > 1, \qquad -1 < x' < 0.$$

Which of the four are eventually periodic?

The test, checked

x	x'	Purely periodic?
$\sqrt{2}$	$-\sqrt{2}$	No
$1+\sqrt{3}$	$1-\sqrt{3}$	Yes
$(1+\sqrt{5})/2$	$(1-\sqrt{5})/2$	Yes
$\sqrt{2}/2$	$-\sqrt{2}/2$	No

All four are eventually periodic.

A visible prefix can belong to the period

$$[2;\overline{1,2}] = 1 + \sqrt{3} = [\overline{2,1}]$$

Its conjugate is $1 - \sqrt{3} \in (-1, 0)$.

Galois detects pure periodicity independently of how we first write the digit string.

Lagrange and Galois

Lagrange

$$\text{eventually periodic} \iff \text{quadratic irrational}$$

Fixed points prove one direction. Bounded integer states prove the other.

Galois

$$\text{purely periodic} \iff x > 1,\ -1 < x' < 0$$

Unique reduced predecessors remove the initial segment.

Exit question

In the proof, where did we use

finiteness?

Where did we use

reversibility?

What can go wrong if only the first property holds?

Choosing the initial integer state

Start with $Ax^2 + Bx + C = 0$, where $A, B, C \in \mathbb{Z}$ and $A \neq 0$.

Change all coefficient signs if needed so that

$$x = \frac{-B + \sqrt{D}}{2A}, \qquad D = B^2 - 4AC > 0.$$

Then choose

$$P_0 = -B, \qquad Q_0 = 2A.$$

$$D - P_0^2 = -4AC = Q_0(-2C).$$

Thus $Q_0 \mid D - P_0^2$. Its sign need not be positive.

The inverse-prefix formula

$$x = \frac{p_{n-1}x_n + p_{n-2}}{q_{n-1}x_n + q_{n-2}}.$$

Solving for x_n gives

$$x_n = -\frac{q_{n-2}x - p_{n-2}}{q_{n-1}x - p_{n-1}}.$$

Conjugating gives

$$x'_n = -\frac{q_{n-2}x' - p_{n-2}}{q_{n-1}x' - p_{n-1}}.$$

For large k, $q_k x' - p_k = q_k(x' - p_k/q_k)$ has a fixed sign.

The conjugate reverses the period

If $x = [\overline{b_0, \ldots, b_{\ell-1}}]$, then

$$-\frac{1}{x'} = [\overline{b_{\ell-1}, \ldots, b_0}].$$

Each digit matrix is symmetric, so reversing a product transposes it.

$$N = \begin{pmatrix} A & B \\ C & D_0 \end{pmatrix}: \quad Cx^2 + (D_0 - A)x - B = 0$$

Substitute $x' = -1/z$:

$$Bz^2 + (D_0 - A)z - C = 0,$$

the fixed-point equation for N^T.

The same matrix produces Pell solutions

Lecture 1 gave

$$U = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}, \qquad T_U(\sqrt{3}) = \sqrt{3}.$$

Ordinary matrix multiplication gives

$$U \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 2p + 3q \\ p + 2q \end{pmatrix}.$$

$$(2p + 3q)^2 - 3(p + 2q)^2 = p^2 - 3q^2.$$

Starting from $(1, 0)$ produces

$$(2, 1),\ (7, 4),\ (26, 15), \ldots$$

Each satisfies $p^2 - 3q^2 = 1$.